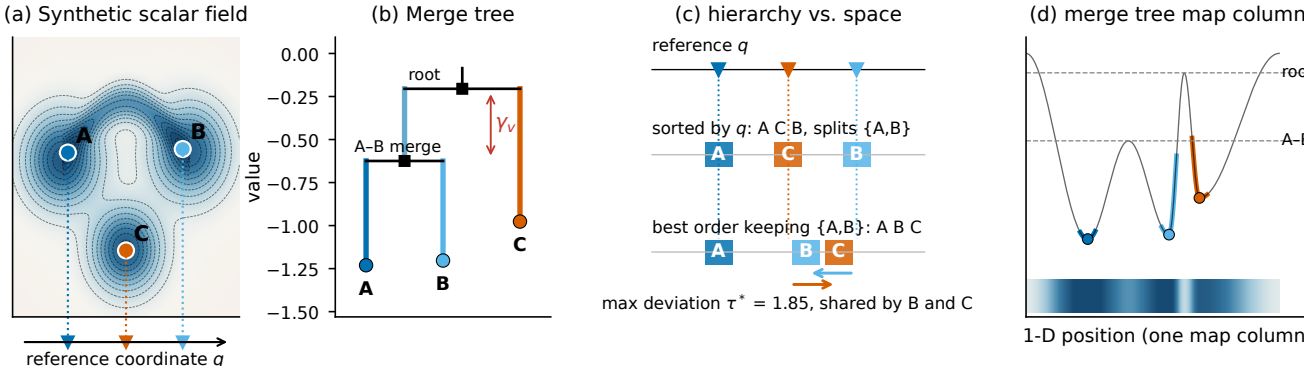

(a) Synthetic scalar field
A
B
C
reference coordinate q
(b) Merge tree
root
A–B merge
Y_v
A
B
C
value
0.00
−0.25
−0.50
−0.75
−1.00
−1.25
−1.50
(c) hierarchy vs. space
reference q
sorted by q: A C B, splits {A,B}
A
C
B
best order keeping {A,B}: A B C
A
B
C
max deviation $\tau^* = 1.85$, shared by B and C
(d) merge tree map column
root
A–B
1-D position (one map column)